University College Cork, Ireland
Coláiste na hOllscoile Corcaigh

Microbiome Ireland

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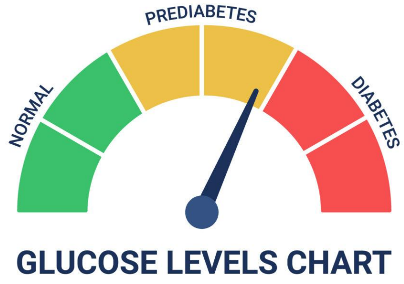
M.Sc. Bioinformatics and Computational Biology

"Single-cell transcriptomics of intestinal epithelial cells: Insights into the prediabetic condition"

Supervised By:
Silvia Melgar

Prediabetes & T2DM



- Global health crisis, T2DM prevalence x4 in last 30 years.
- Characterised by: insulin resistance, inadequate insulin secretion, leads to hyperglycaemia.
- Usually preceded by prediabetes. (Impaired fasting glucose, Impaired glucose tolerance, raised HbA1c levels).
- Associated with: obesity, diet, sedentary lifestyle, genetic factors.
- Systemic effects of T2DM and prediabetes are well documented, although emerging evidence is highlighting the role of the intestinal epithelium in the development of prediabetes.

Project Aim

Find altered genes/pathways in the individual intestinal cell types which are characteristic of the prediabetic subject.

Why?

Complement of another study

Allows for the selection of biological targets for further study of seaweed extract.

Intestinal Epithelium

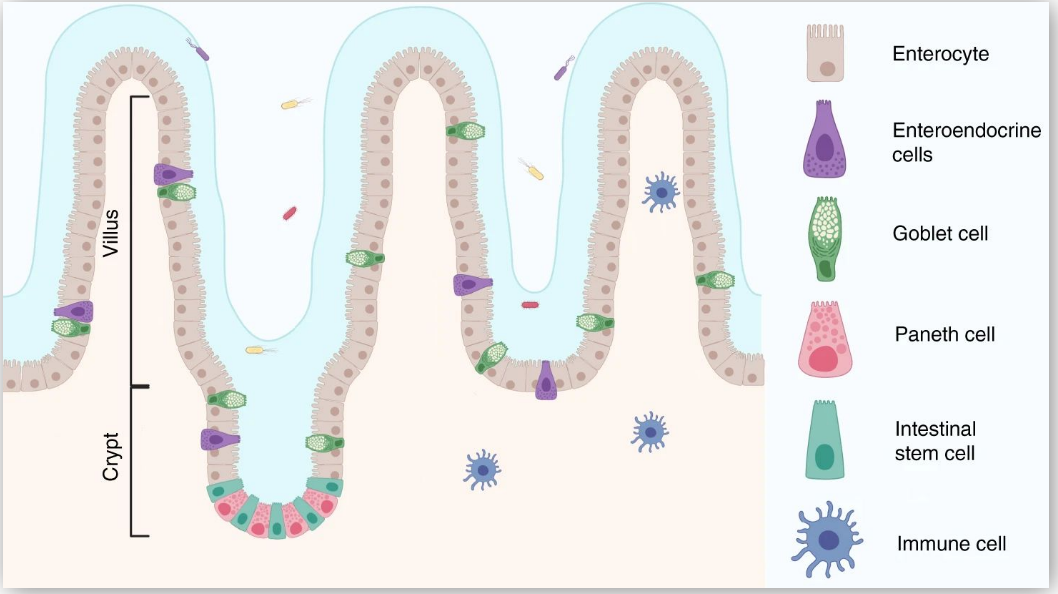


Image adapted from: https://www.nature.com/articles/s41467-020-20052-z/figures/1

How?

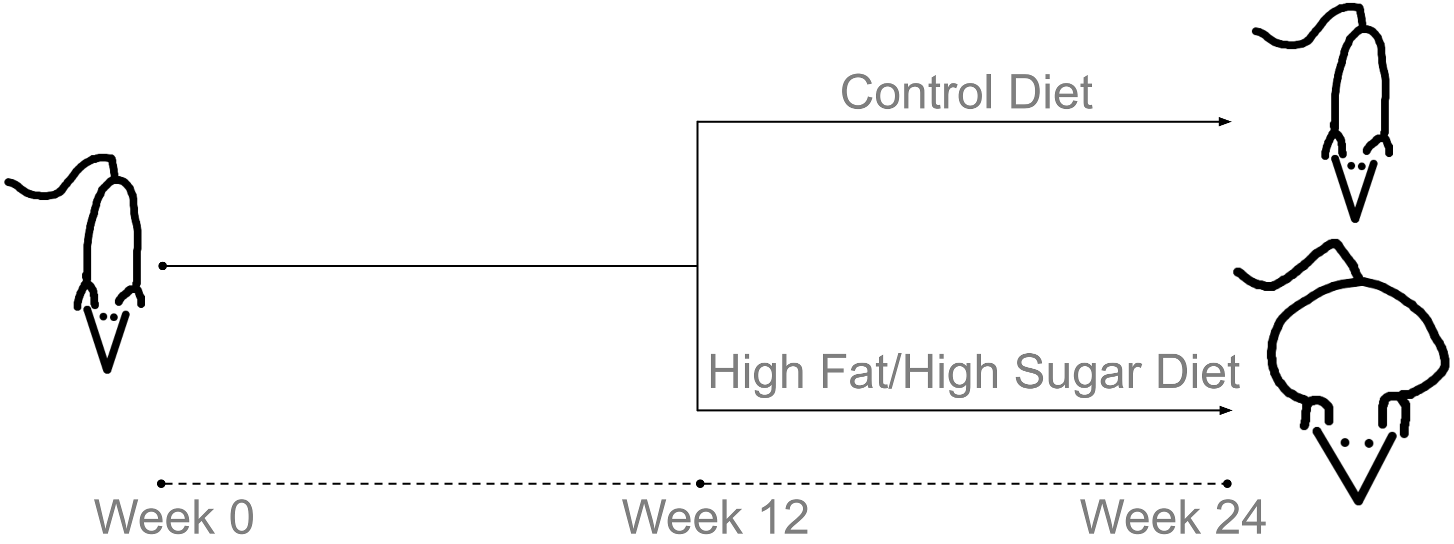
- Choose suitable data
- Review literature for best analysis practices
- Build an analysis pipeline
- Carry out the analysis
- Review the findings
- Form a biological hypothesis
- Build onto the current literature

Experimental Model


ARTICLES
https://doi.org/10.1038/s42255-021-00458-9
nature metabolism
Check for updates
OPEN
Diet-induced alteration of intestinal stem cell function underlies obesity and prediabetes in mice

https://www.nature.com/articles/s42255-021-00458-9

- Mice assigned to one of two diets for 12 weeks, until 24 weeks old.
- Prediabetic status evaluated.
- Intestinal tissue taken and sequenced at the single cell resolution.



The Upstream Pipeline

A series of automated, sequential steps for processing and analysing biological data, ensuring efficient and reproducible workflows.

Raw Data (Count Table) → Data Cleaning → Quality Control → Doublet Removal

↓

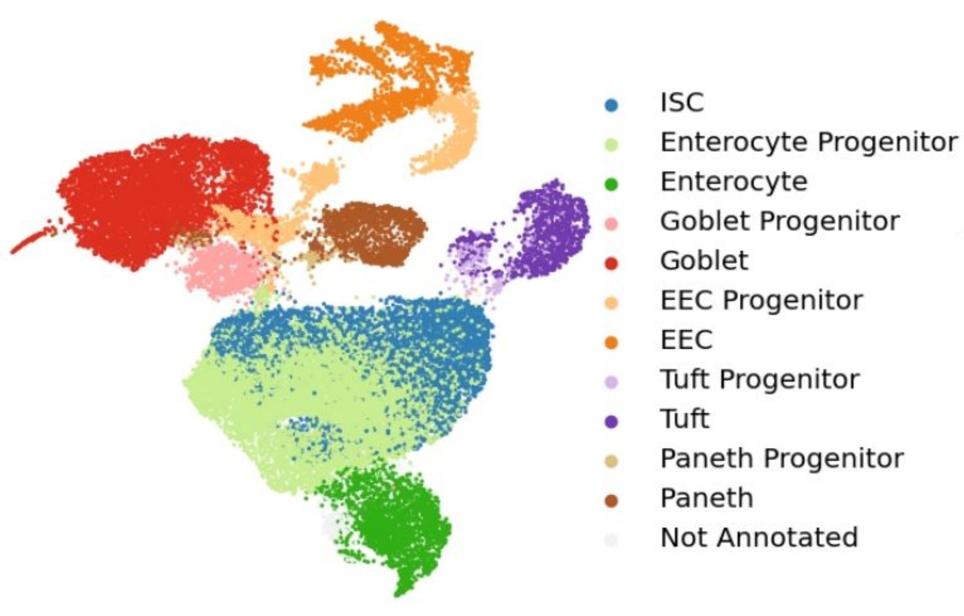
Normalisation → Feature Selection → PCA → kNN/UMAP

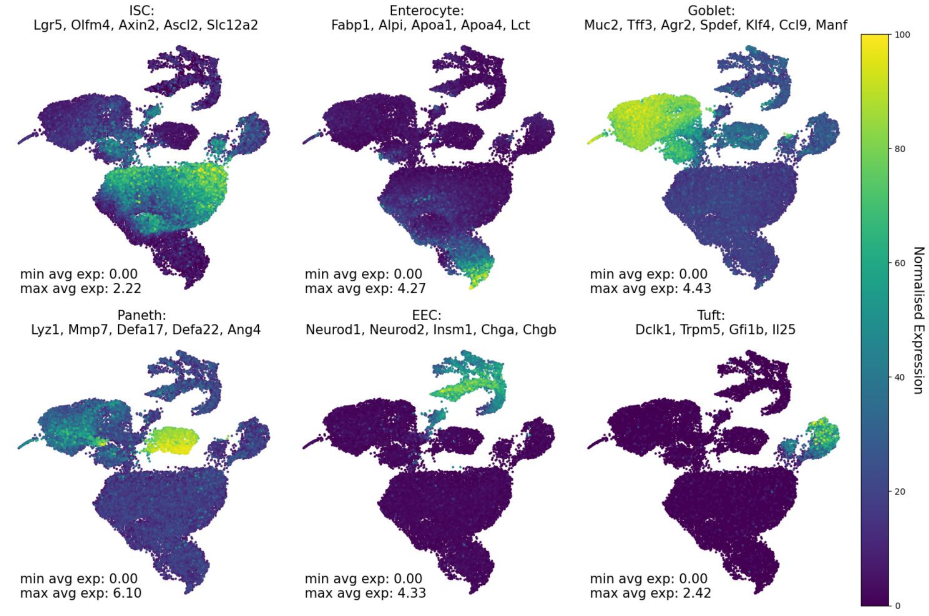
↓

Clustering → Cell Annotation

Annotated data table ✅

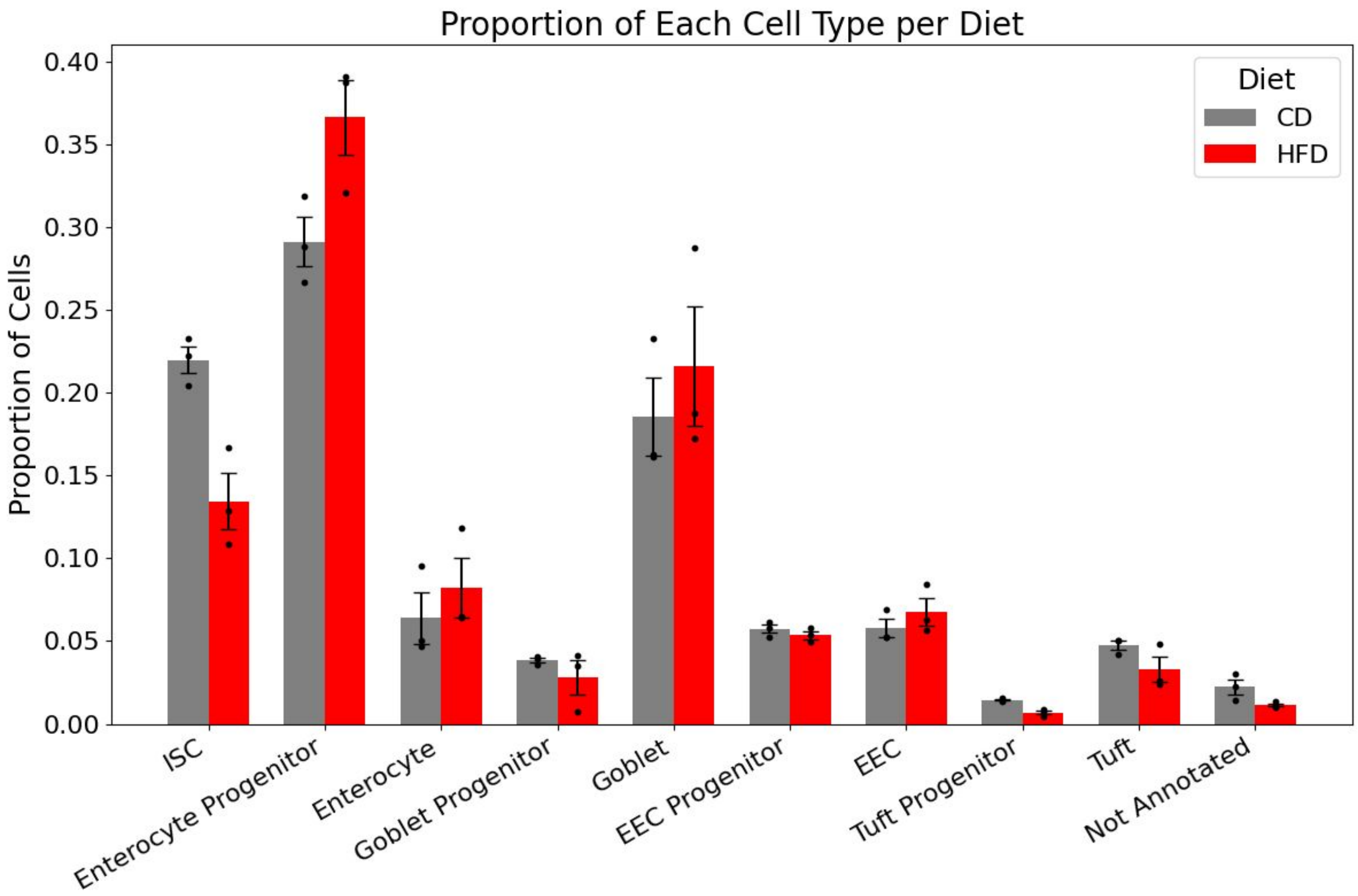
Annotated dataset



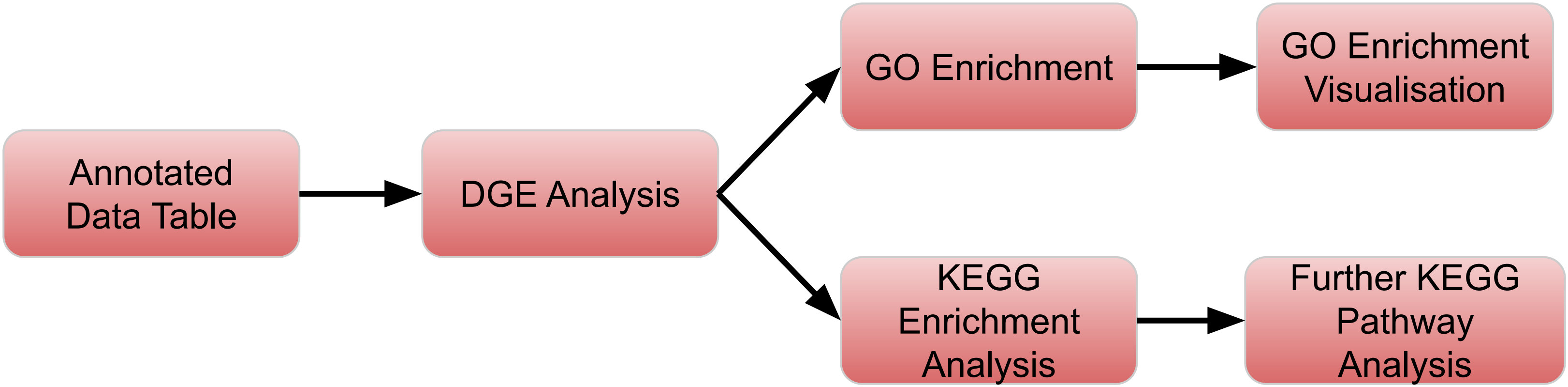


- 27,419 cells (13,363 Control Diet, 14,056 High-fat high-sugar Diet)
- 6 main cell types, with respective progenitor cell types.

Cell Type Proportions



The Downstream Pipeline



Abbreviations:
DGE: Differential Gene Expression
GO: Gene Ontology
KEGG: Kyoto Encyclopedia of Genes and Genomes

Gene Set Enrichment - Gene Ontology

GO Enrichment - What did we find?

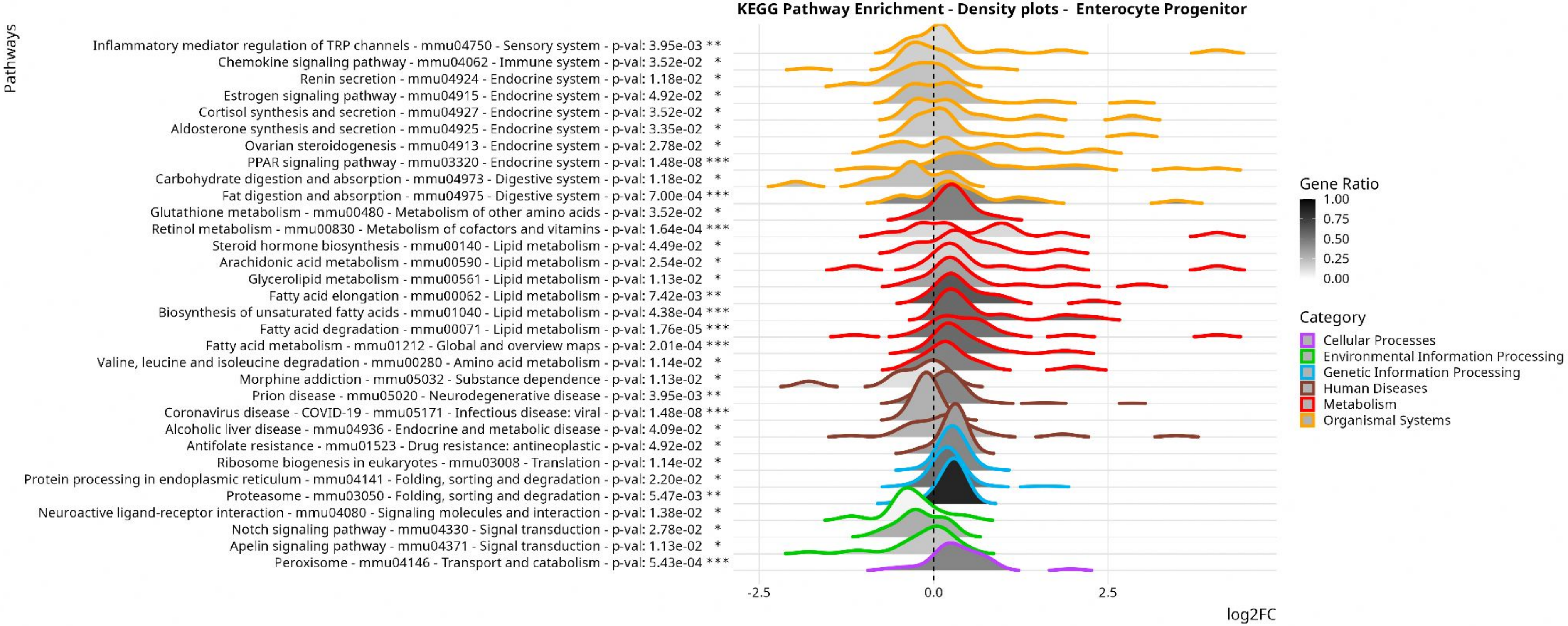
Intestinal Stem Cells

- ↑ Cell cycle, chromosome dynamics, and DNA replication processes across biological processes, cellular components, and molecular functions.
- ↑ Mitochondrial function
- ↓ Adherens junctions (paracellular barrier)

Enterocytes

- ↑ lipid metabolism processes
- ↓ cell adhesion and signalling (paracellular barrier)
- ↓ calcium-dependent protein binding

Gene Set Enrichment



KEGG Enrichment - What did we find?

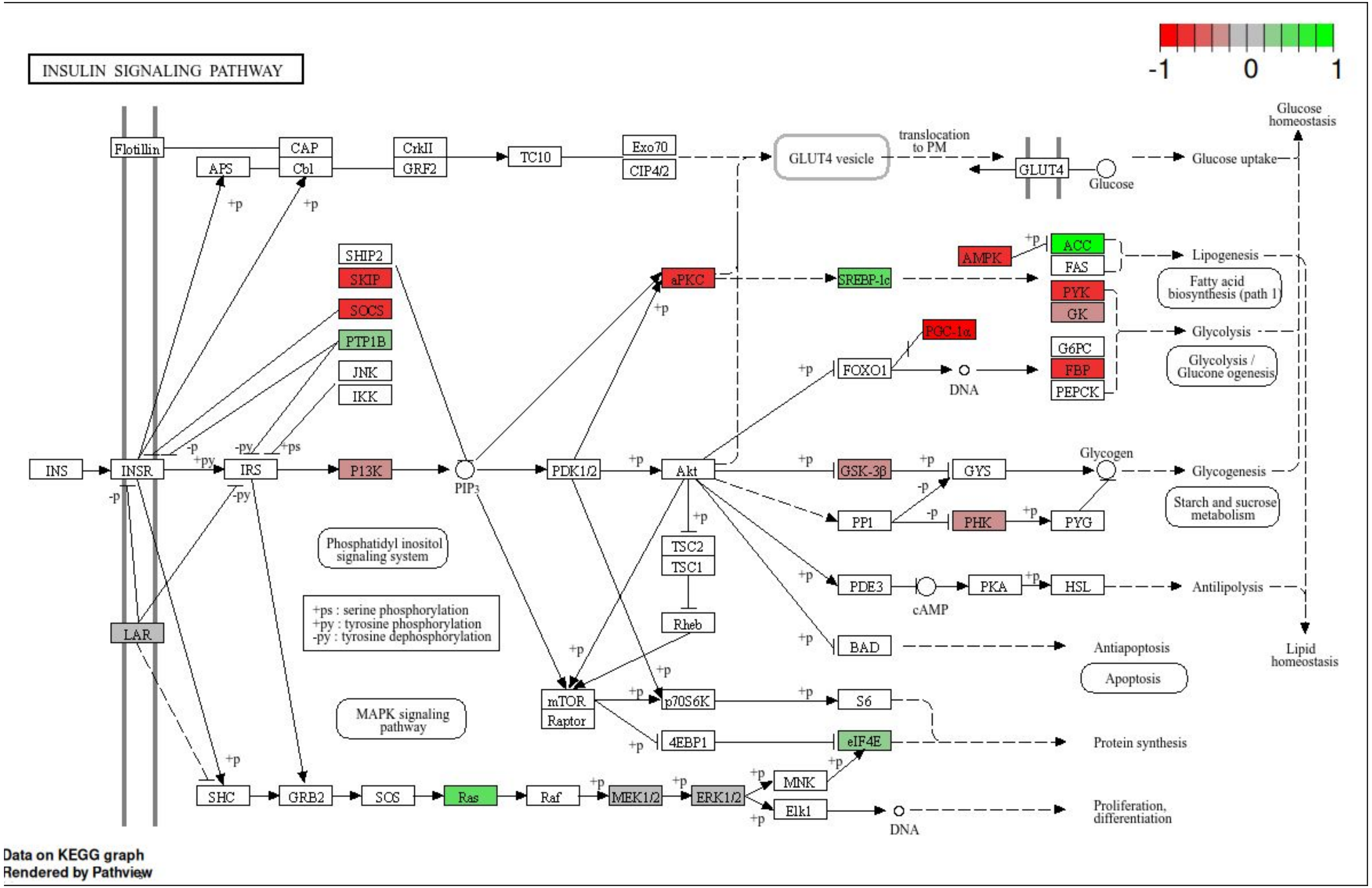
Intestinal Stem Cells

- ↑ Cell Cycle
- ↓ Cell adhesion molecules pathway (Paracellular Junction)
- ↑ Fatty acid metabolism pathway

Enterocytes

- ↓ Carbohydrate Digestion and absorption
- ↑ Lipid metabolism and digestion pathways
- ↑ Protein processing in the endoplasmic reticulum
- ↑ Proteasome

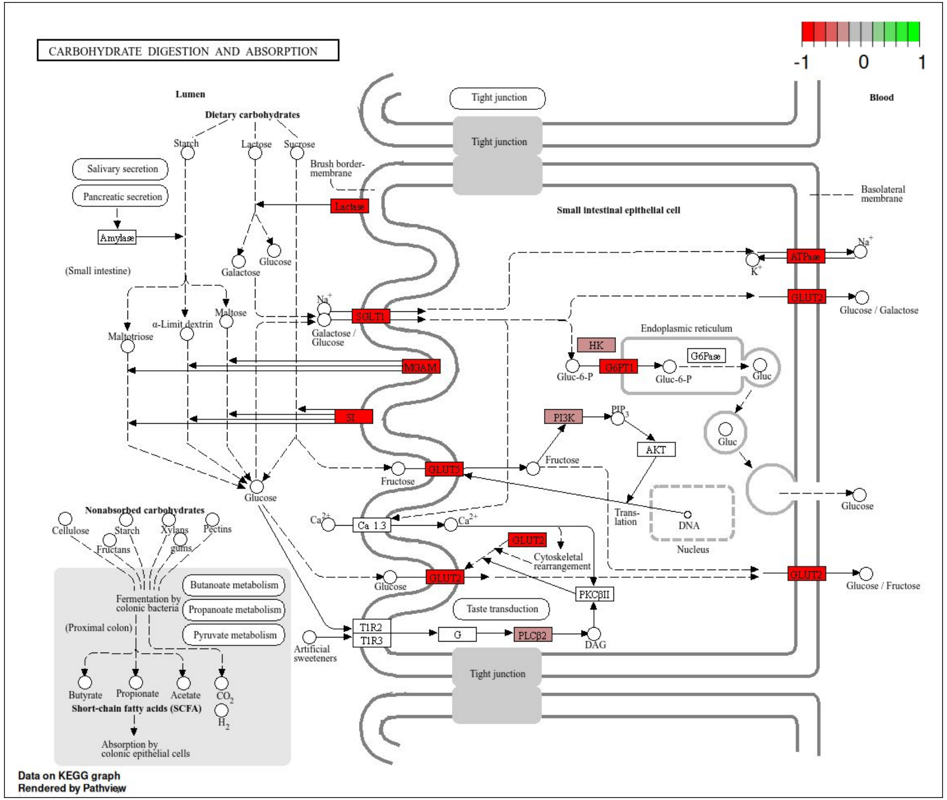
HFHSD alters macronutrient metabolism in the IE layer

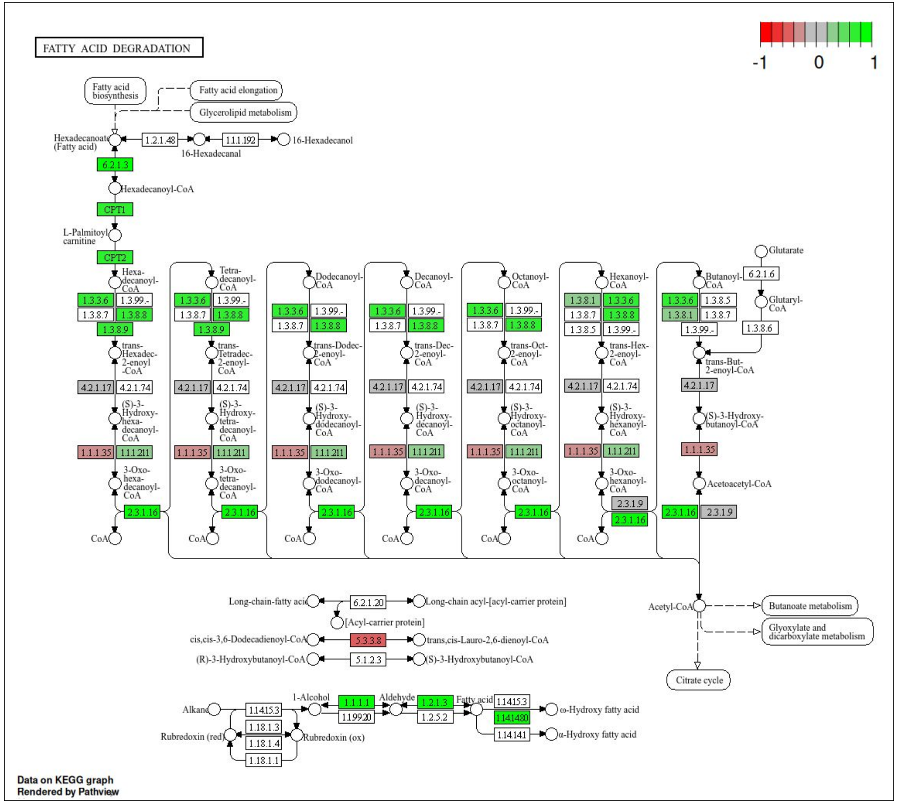


- **Srebp1-c** transcriptional regulator is upregulated
- Involved in lipid metabolism and glycolysis

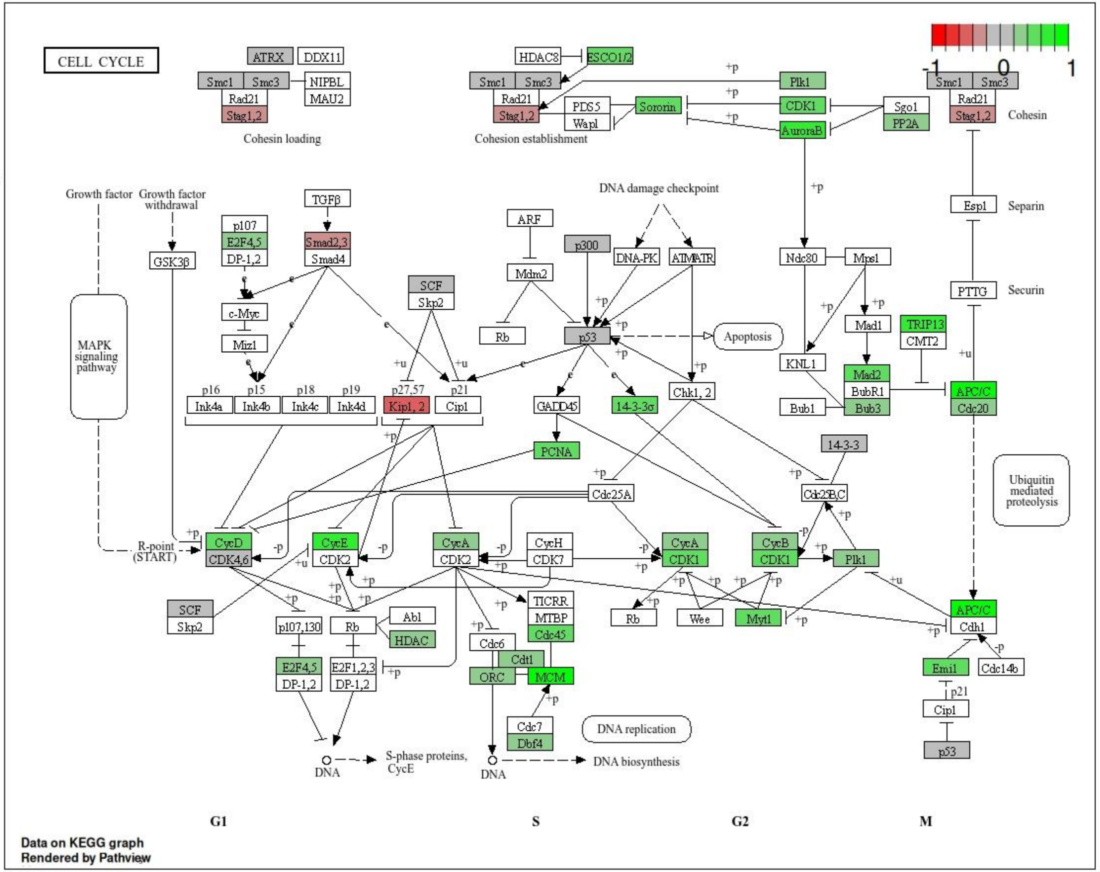
Carbohydrate / Fatty Acid Metabolism alterations

- Enterocytes shift from carbohydrate utilisation to lipid utilisation





HFHSD induces hyperproliferation through upregulation of the cell cycle.

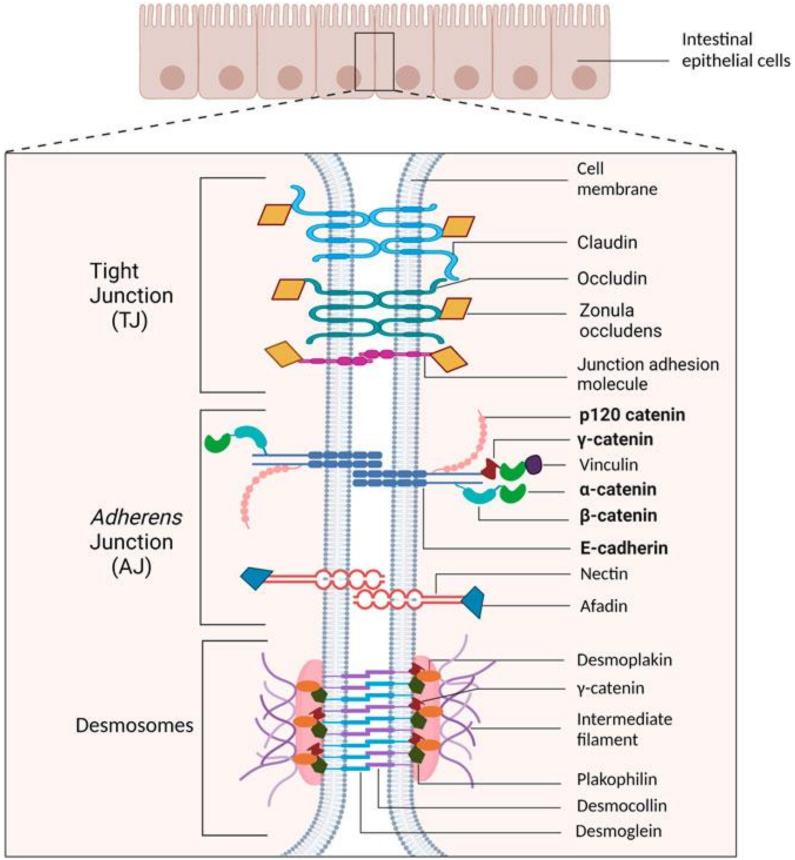


Barrier Permeability & Inflammation

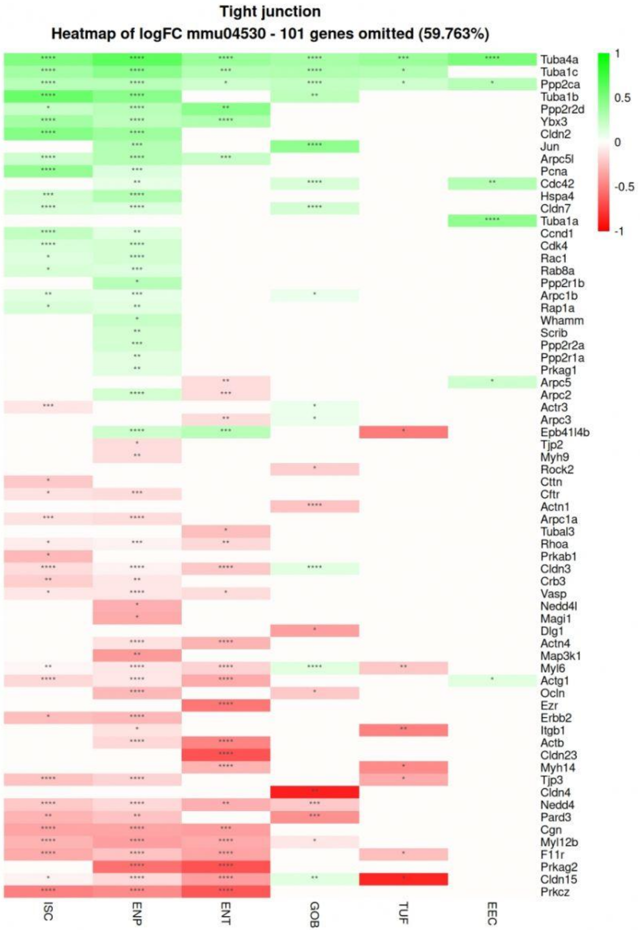
- Systemic inflammation in prediabetes is characterised by elevated levels of inflammatory markers and alterations in immune function

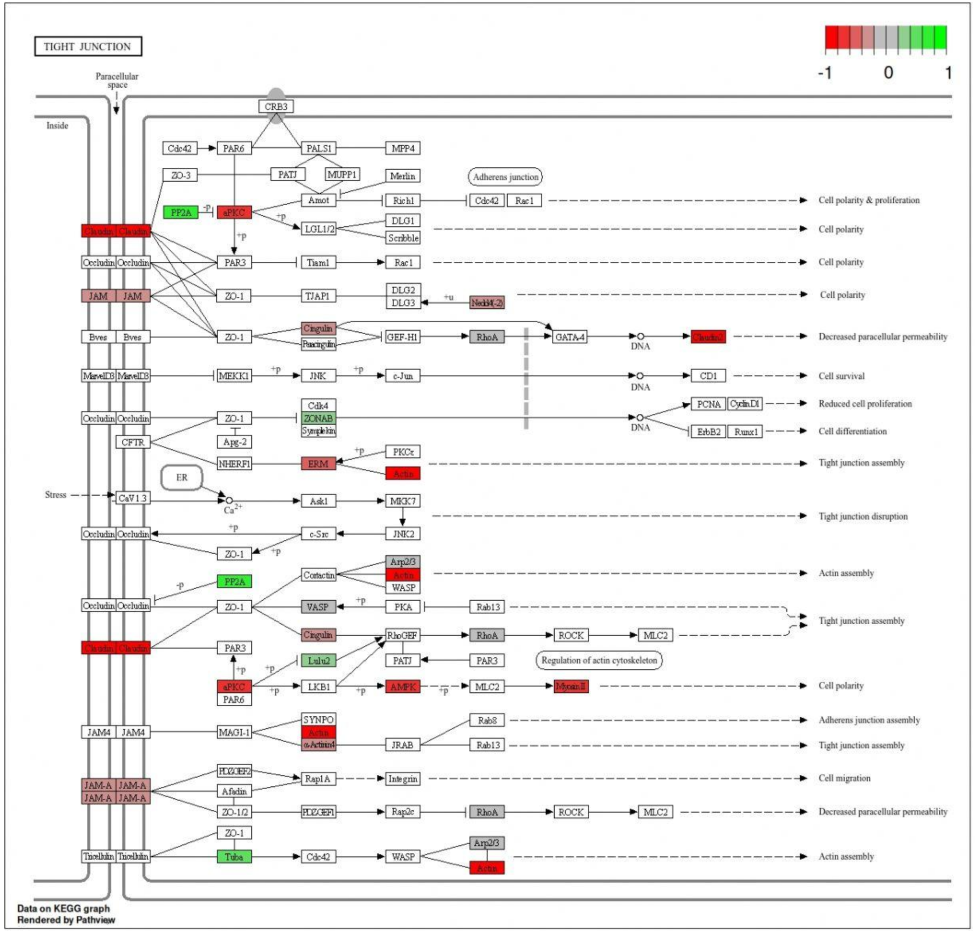
- Inflammation during the prediabetic state seems to be a driving force behind pancreatic beta cell dysfunction, insulin resistance, dyslipidemia, and cardiovascular diseases and is a risk factor for peripheral vascular diseases

- Emerging evidence is linking ‘leaky gut’ to inflammation, we decided to test this hypothesis in an attempt to clarify conflicts in the literature.



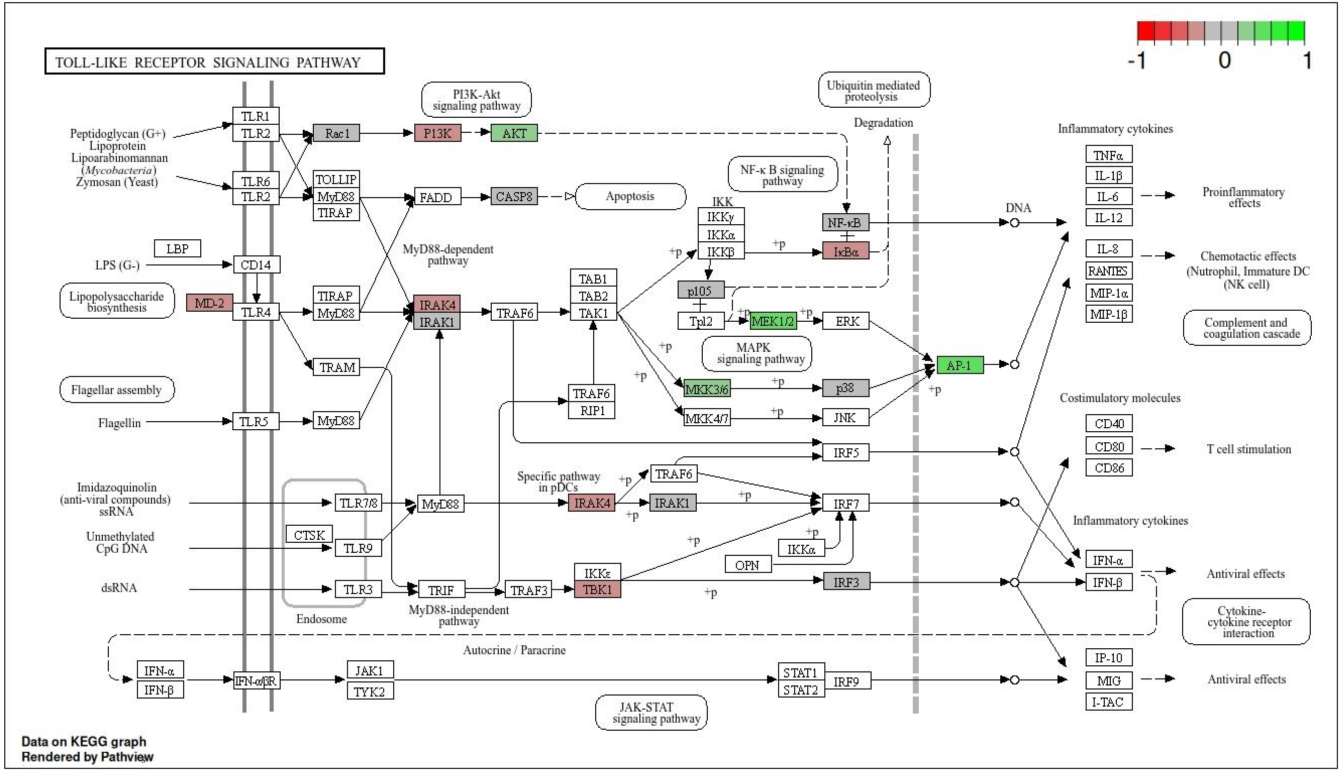
Intestinal Barrier Integrity





Inflammation

- Detection of inflammation was <u>inconclusive</u>



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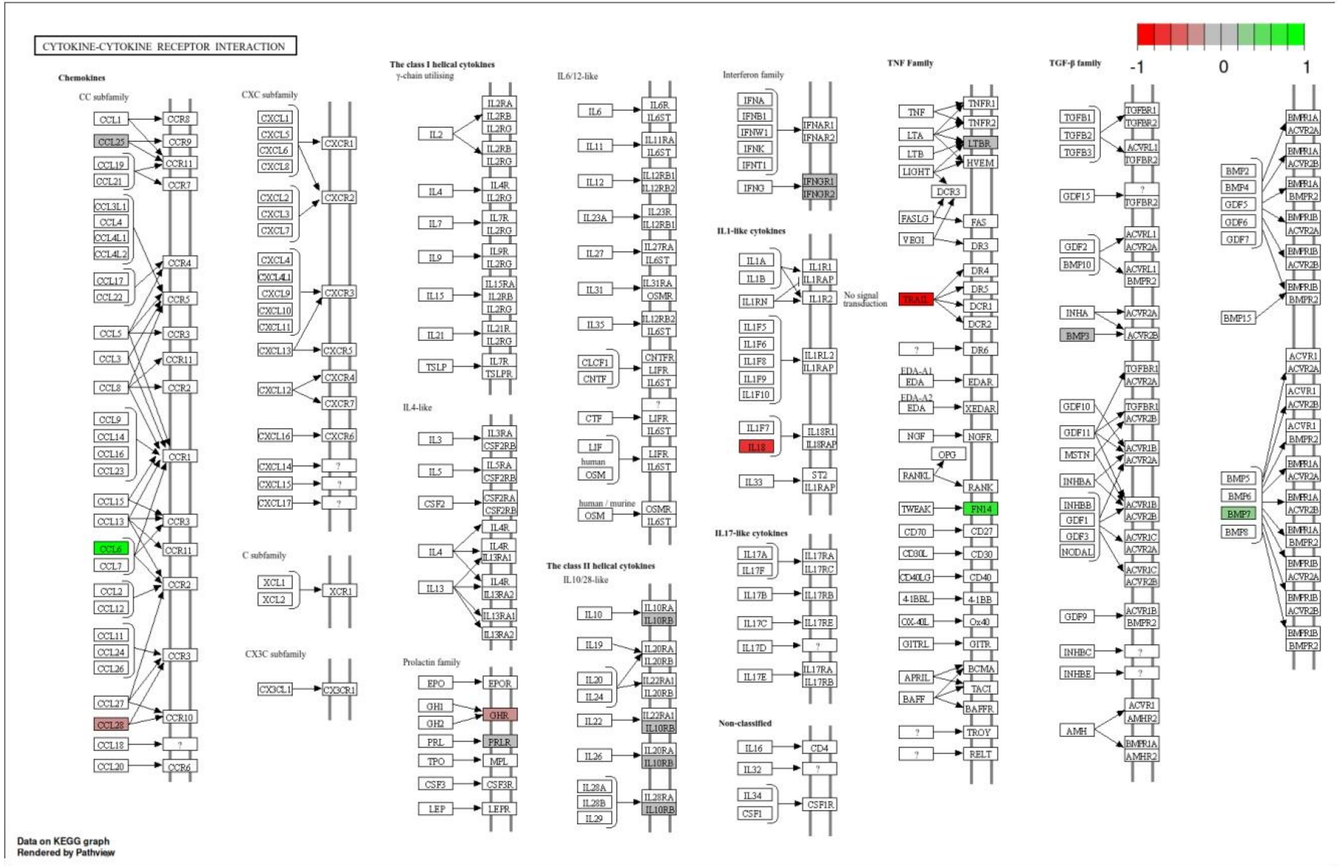
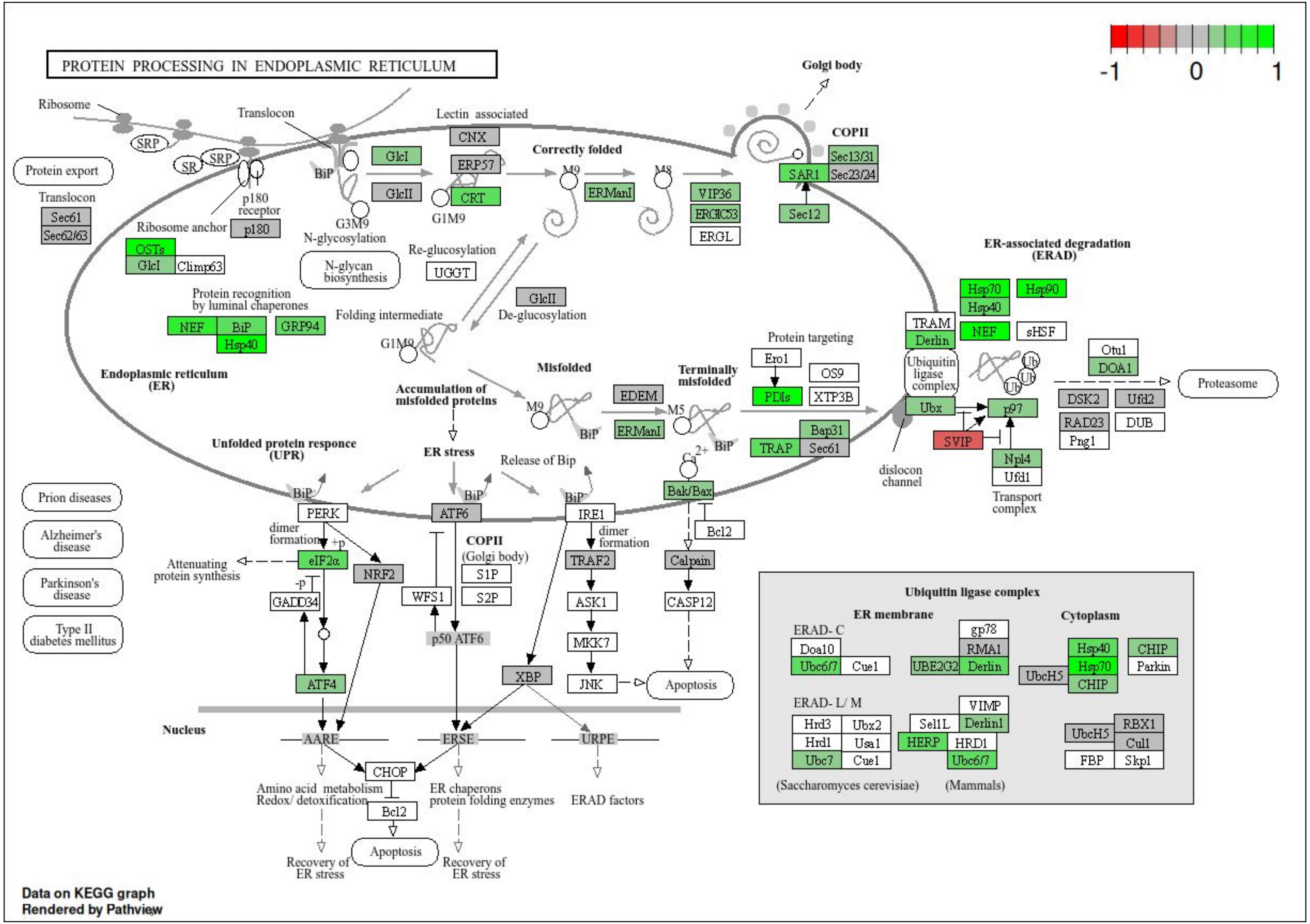




ER Stress

ER stress is induced by three pathways: **PERK**, **ATF6**, **IRE1**

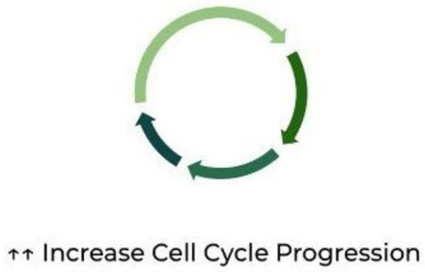
We detected an upregulation of the PERK pathway.

Widespread upregulation of ER Stress pathways.

Perhaps helps to correct impaired intestinal barrier.

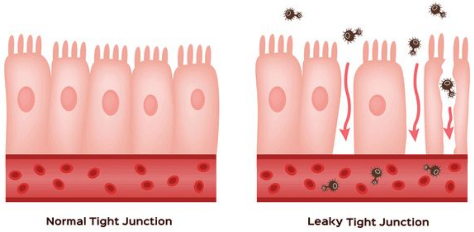
Conclusions

Intestinal Stem Cell - Cell Cycle is increased resulting in faster turnover of the IE layer

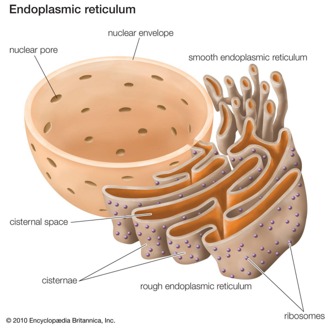


Carbs → Fats

Enterocytes shift from carbohydrate utilisation to fat utilisation for energy.



Intestinal Barrier Integrity is impaired although there is no evidence to support that this leads to inflammation.



ER stress is induced by HFHSD perhaps to deal with

Thank you!

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